

Questions

MathonGo

Q1 - 24 June - Shift 1

Two identical cells each of emf 1.5 V are connected in parallel across a parallel combination of two resistors each of resistance 20Ω . A voltmeter connected in the circuit measures 1.2 V.

The internal resistance of each cell is

(A) 2.5Ω (B) 4Ω

(C) 5Ω (D) 10Ω

Space for your notes:

Q2 - 24 June - Shift 1

In a potentiometer arrangement, a cell gives a balancing point at 75 cm length of wire. This cell is now replaced by another cell of unknown emf. If the ratio of the emf's of two cells respectively is 3 : 2, the difference in the balancing length of the potentiometer wire in above two cases will be _____ cm.

Space for your notes:

Q3 - 24 June - Shift 2

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What will be the most suitable combination of three resistors $A = 2\Omega$, $B = 4\Omega$, $C = 6\Omega$ so that $\left(\frac{22}{3}\right)\Omega$ is equivalent resistance of combination?

Space for your notes:

- (A) Parallel combination of A and C connected in series with B.
- (B) Parallel combination of A and B connected in series with C.
- (C) Series combination of A and C connected in parallel with B.
- (D) Series combination of B and C connected in parallel with A.

Q4 - 24 June - Shift 2

A potentiometer wire of length 10 m and resistance 20Ω is connected in series with a 25 V battery and an external resistance 30Ω . A cell of emf E in secondary circuit is balanced by 250 cm long potentiometer wire. The value of E (in volt) is $\frac{x}{10}$. The value of x is _____.

*Space for your notes:***Q5 - 25 June - Shift 1**

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A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the below is **true** for measuring value of G ?

Space for your notes:

- (A) $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.
- (B) $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistance(s).
- (C) $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistance(s).
- (D) $\frac{1}{3}$ deflection method can be used and in this case the G value equals to the shunt resistance(s).

Q6 - 25 June - Shift 1

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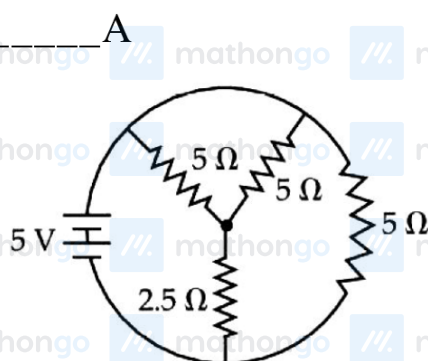
Questions

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A resistor develops 300 J of thermal energy in 15s, when a current of 2A is passed through it. If the current increases to 3A, the energy developed in 10s is _____ J.

*Space for your notes:***Q7 - 25 June - Shift 1**

The total current supplied to the circuit as shown in figure by the 5V battery is

Space for your notes:**Q8 - 25 June - Shift 2**

Two cells of same emf but different internal resistances r_1 and r_2 are connected in series with a resistance R . The value of resistance R , for which the potential difference across second cell is zero, is

Space for your notes:

(A) $r_2 - r_1$

(B) $r_1 - r_2$

(C) r_1

(D) r_2

Q9 - 25 June - Shift 2

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If n represents the actual number of deflections in a converted galvanometer of resistance G and shunt resistance S . Then the total current I when its figure of merit is K will be :

- (A) $\frac{KS}{(S+G)}$ (B) $\frac{(G+S)}{nKS}$
(C) $\frac{nKS}{(G+S)}$ (D) $\frac{nK(G+S)}{S}$

Space for your notes:

Q10 - 25 June - Shift 2

The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be _____%.

Space for your notes:

Q11 - 26 June - Shift 1

An aluminium wire is stretched to make its length, 04% larger. Then percentage change in resistance is:

- (A) 0.4 % (B) 0.2 %
(C) 0.8 % (D) 0.6 %

Space for your notes:

Q12 - 26 June - Shift 2

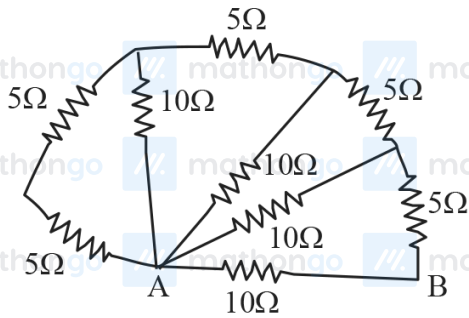
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The equivalent resistance between points A and B in the given network is :

Space for your notes:



- (A) 65Ω (B) 20Ω
(C) 5Ω (D) 2Ω

Q13 - 27 June - Shift 1

A 72Ω galvanometer is shunted by a resistance of 8Ω . The percentage of the total current which passes through the galvanometer is :

Space for your notes:

- (A) 0.1% (B) 10 %
(C) 25% (D) 0.25%

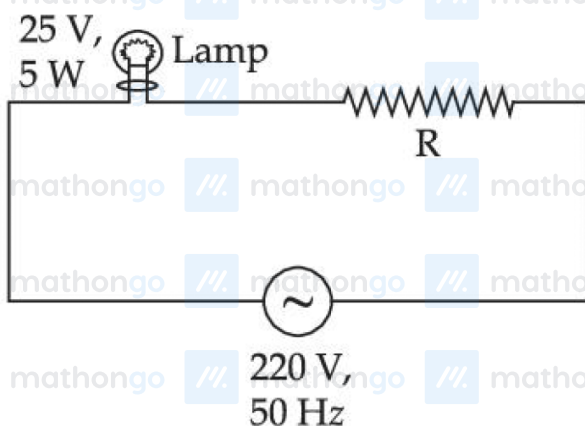
Q14 - 27 June - Shift 1

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A 220 V, 50 Hz AC source is connected to a 25 V, 5 W lamp and an additional resistance R in series (as shown in figure) to run the lamp at its peak brightness, then the value of R (in ohm) will be

Space for your notes:**Q15 - 27 June - Shift 1**

A cell, shunted by a $8\ \Omega$ resistance, is balanced across a potentiometer wire of length 3m. The balancing length is 2 m when the cell is shunted by $4\ \Omega$ resistance. The value of internal resistance of the cell will be $\underline{\hspace{1cm}}\ \Omega$.

*Space for your notes:***Q16 - 27 June - Shift 1**

The current density in a cylindrical wire of radius 4 mm is $4 \times 10^6\ \text{A m}^{-2}$. The current through the outer portion of the wire between radial distance $\frac{R}{2}$ and R is $\underline{\hspace{1cm}}\ \pi\ \text{A}$.

*Space for your notes:***Q17 - 27 June - Shift 2**

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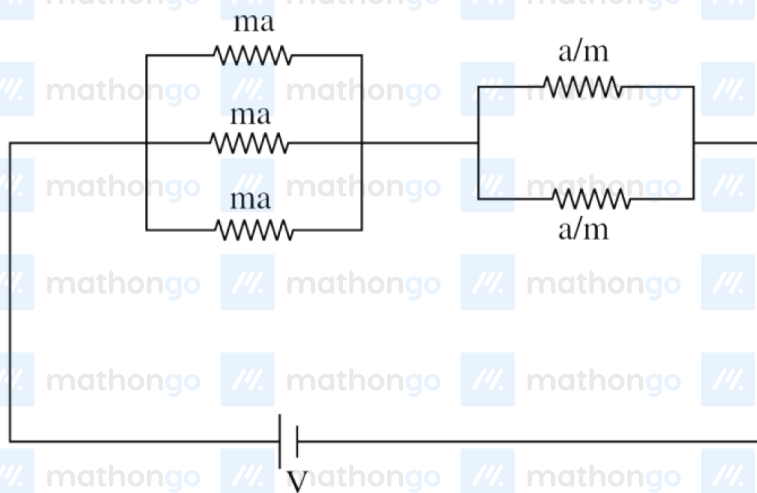
The current density in a cylindrical wire of radius $r = 4.0 \text{ mm}$ is $1.0 \times 10^6 \text{ A/m}^2$. The current through the outer portion of the wire between radial distances $r/2$ and r is $x\pi \text{ A}$; where x is _____.

Space for your notes:

Q18 - 27 June - Shift 2

In the given circuit 'a' is an arbitrary constant. The value of m for which the equivalent circuit resistance is minimum, will be $\sqrt{\frac{x}{2}}$. The value of x is _____.

Space for your notes:



Q19 - 28 June - Shift 1

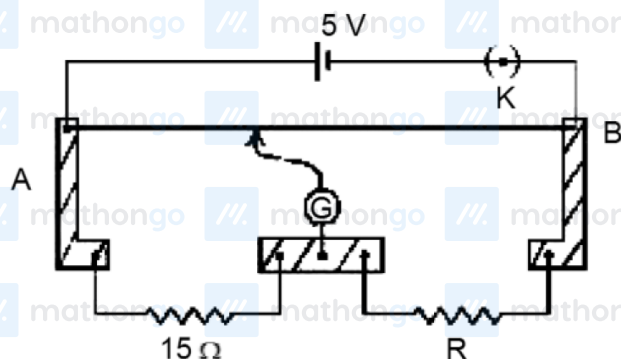
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A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of $15\ \Omega$. The galvanometer (G) shows null deflection when tapping key is at 43 cm mark from end A. If the end correction for end A is 2 cm, then the determined value of R will be _____ Ω .

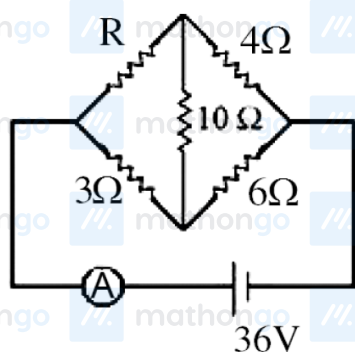
Space for your notes:



Q20 - 28 June - Shift 1

Current measured by the ammeter (A) in the reported circuit when no current flows through $10\ \Omega$ resistance, will be _____ A.

Space for your notes:



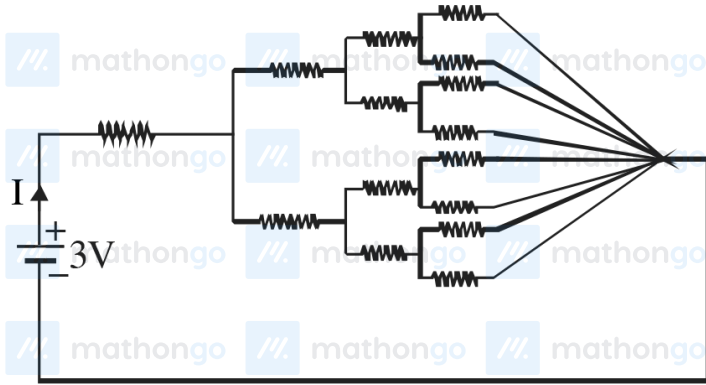
Q21 - 28 June - Shift 2

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All resistances in figure are 1Ω each. The value of current 'I' is $\frac{a}{5}$ A. The value of a is _____

Space for your notes:**Q22 - 29 June - Shift 1**

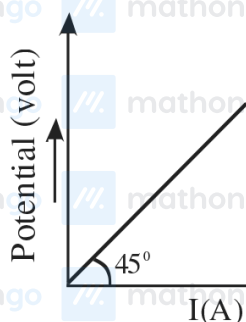
Two coils require 20 minutes and 60 minutes respectively to produce same amount of heat energy when connected separately to the same source. If they are connected in parallel arrangement to the same source; the time required to produce same amount of heat by the combination of coils, will be _____ min.

*Space for your notes:***Q23 - 29 June - Shift 1**

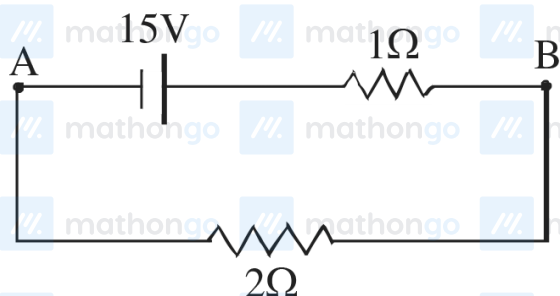
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The variation of applied potential and current flowing through a given wire is shown in figure. The length of wire is 31.4 cm. The diameter of wire is measured as 2.4 cm. The resistivity of the given wire is measured as $x \times 10^{-3} \Omega \text{ cm}$. The value of x is _____. [Take $\pi = 3.14$]

*Space for your notes:***Q24 - 29 June - Shift 1**

For the network shown below, the value $V_B - V_A$ is _____ V.

*Space for your notes:***Q25 - 29 June - Shift 2**

Questions

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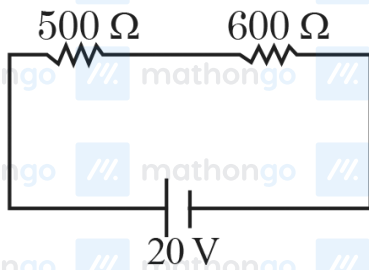
The combination of two identical cells, whether connected in series or parallel combination provides the same current through an external resistance of 2Ω . The value of internal resistance of each cell is :

Space for your notes:

- (A) 2Ω (B) 4Ω (C) 6Ω (D) 8Ω

Q26 - 29 June - Shift 2

Two resistors are connected in series across a battery as shown in figure. If a voltmeter of resistance 2000Ω is used to measure the potential difference across 500Ω resistor, the reading of the voltmeter will be ____ V.

Space for your notes:

Answer Key

Q1 (C)

Q2 (25)

Q3 (B)

Q4 (25)

Q5 (B)

Q6 (450)

Q7 (2)

Q8 (A)

Q9 (D)

Q10 (300)

Q11 (C)

Q12 (C)

Q13 (B)

Q14 (975)

Q15 (8)

Q16 (48)

Q17 (12)

Q18 (3)

Q19 (19)

Q20 (10)

Q21 (8)

Q22 (15)

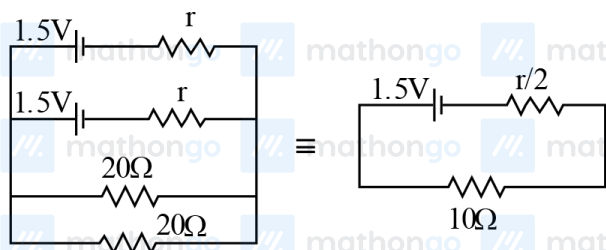
Q23 (144)

Q24 (10)

Q25 (A)

Q26 (8)

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Q1 (C)

$$V = E - ir/2$$

$$1.2 = 1.5 - i \left(\frac{r}{2} \right)$$

$$i \frac{r}{2} = 0.3$$

$$i = \frac{1.5}{10 + \frac{r}{2}} \Rightarrow 10i + \frac{ir}{2} = 1.5$$

$$10i = 1.5 - 0.3$$

$$i = 0.12 \text{ A}$$

$$\Rightarrow r = \frac{0.6}{0.12} = 5\Omega$$

Q2 (25)

$$\frac{\epsilon_1}{\epsilon_2} = \frac{l_1}{l_2}$$

$$\frac{3}{2} = \frac{75\text{cm}}{l_2}$$

$$l_2 = 50\text{cm}$$

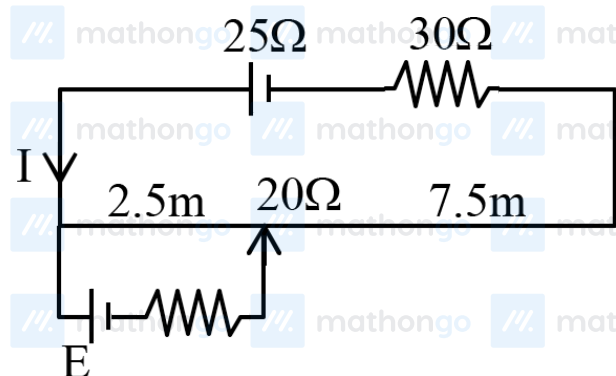
$$l_1 - l_2 = 75 - 50$$

$$= 25\text{ cm}$$

Q3 (B)

$$\Rightarrow \frac{4}{3} + 6 = \frac{22}{3}$$

Q4 (25)



$$I = \frac{25}{50} = \frac{1}{2} \text{ A}$$

$$\therefore \Delta V = 10 \text{ V}$$

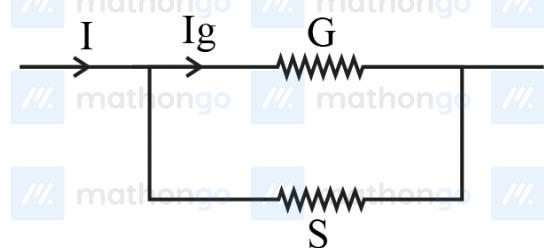
$$10 \text{ m} \rightarrow 10 \text{ V}$$

$$2.5 \text{ m} \rightarrow 2.5 \text{ V}$$

Q5 (B)

In galvanometer

$$\Rightarrow (I - I_g)S = I_g G$$



$$\frac{I_g}{I} = \frac{S}{S + G}$$

$$\Rightarrow \frac{1}{3} = \frac{S}{S + G} \Rightarrow S + G = 3S \Rightarrow G = 2S$$

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Q6 (450)

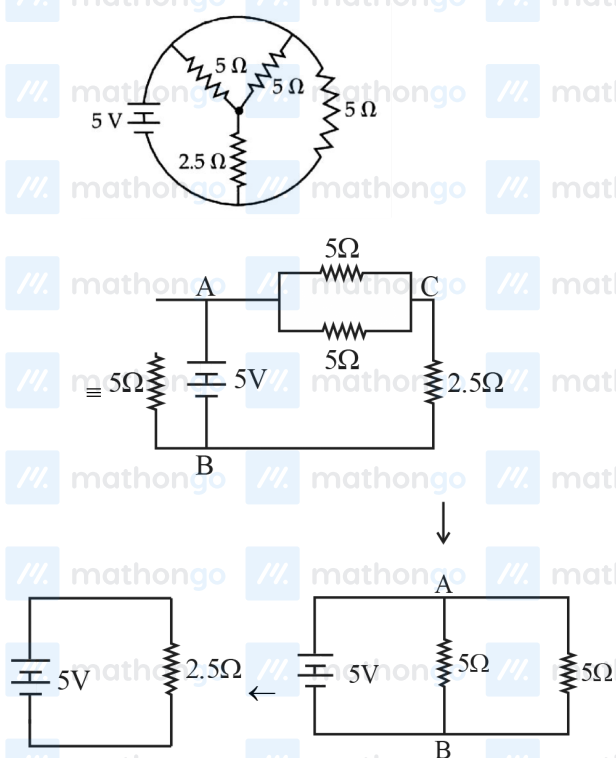
$$H = i^2 R t$$

$$300 = 2^2 \times R \times 15$$

$$\Rightarrow R = \frac{300}{60} = 5\Omega$$

Now, for $i = 3\text{A}$, $t = 10\text{s}$, $R = 5\Omega$

$$H = 3^2 \times 5 \times 10 = 450 \text{ J}$$

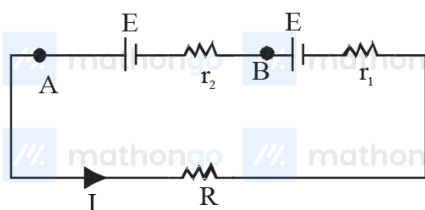
Q7 (2)

Current supplied by 5V battery

$$= \frac{5\text{V}}{2.5\Omega} = 2\text{A}$$

Q8 (A)

$$I = \frac{2E}{R + r_1 + r_2} \dots (i)$$



But $V_A - V_B = E - Ir_2 = 0$

$$\Rightarrow I = \frac{E}{r_2} \dots (ii)$$

Comparing values of I from (i) and (ii)

$$\frac{E}{r_2} = \frac{2E}{R + r_1 + r_2}$$

$$\Rightarrow R = r_2 - r_1$$

Q9 (D)

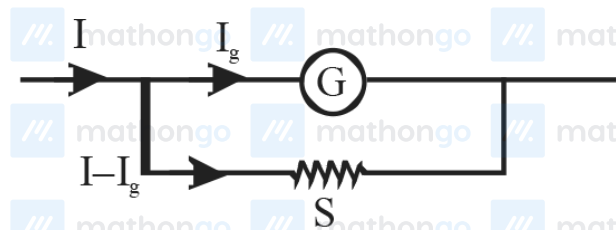


Figure of merit $\frac{I_g}{\theta} = K$

$$I_g = Kn$$

$$I = \frac{I_g}{S} (G + S)$$

$$I = \frac{nK}{S} (G + S)$$

Q10 (300)

$$V' = V$$

$$\ell'A = \ell A$$

$$2\ell A' = \ell A$$

$$A' = \frac{A}{2}$$

$$R = \rho \frac{\ell}{A} \dots (i)$$

$$\ell' = 2\ell$$

$$A' = \frac{A}{2}$$

$$R' = \frac{\rho \ell'}{A'} = \frac{\rho 2\ell}{\frac{A}{2}}$$

$$R' = \frac{4\rho \ell}{A}$$

$$R' = 4R \text{ from equation (i)}$$

% increase in resistance

$$= \frac{R' - R}{R} \times 100 = \frac{4R - R}{R} \times 100$$

$$= 300 \%$$

Q11 (C)

$$R = \frac{\rho l}{A}$$

$$\frac{\Delta R}{R} = \frac{\Delta l}{l} + \frac{\Delta A}{A}$$

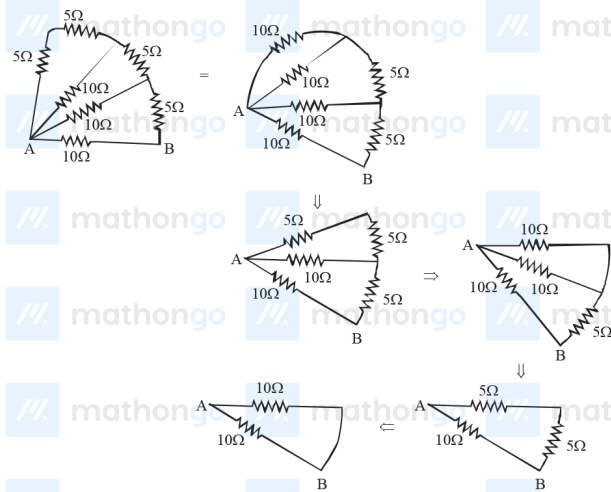
$$lA = k$$

$$\frac{\Delta l}{l} + \frac{\Delta A}{A} = 0$$

$$\frac{\Delta R}{R} = \frac{2\Delta l}{l}$$

$$\frac{\Delta R}{R} = 2 \times 0.4 = 0.8\%$$

Q12 (C)



$$R_{AB} = 5\Omega$$

Q13 (B)

$$S = \frac{R_G}{\frac{I}{I_g} - 1}$$

$$8 = \frac{72}{\frac{I}{I_g} - 1}$$

$$\frac{I}{I_g} - 1 = 9$$

$$\frac{I}{I_g} = 10 \Rightarrow \frac{I_g}{I} = \frac{1}{10} \quad \% I = \frac{I_g}{I} \times 100 = 10\%$$

Q14 (975)

$$P = Vi$$

$$5 = 25i$$

$$i = \frac{1}{5}$$

$$V_R = iR$$

$$(220 - 25) = \frac{1}{5} R$$

$$R = 195 \times 5 = 975 \Omega$$

Q15 (8)

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$$\frac{V_1}{V_2} = \frac{3}{2} = \frac{E - i_1 r}{E - i_2 r}$$

$$= \frac{E - \frac{E}{8 + r} \times r}{E - \frac{E}{4 + r} \times r}$$

$$\frac{3}{2} = \frac{8(4 + r)}{4(8 + r)}$$

$$24 + 3r = 16 + 4r$$

$$r = 8\Omega$$

Q16 (48)

$$J = \frac{I}{A}$$

$$I = JA$$

$$= 4 \times 10^6 \times \left[\pi R^2 - \pi \left(\frac{R}{2} \right)^2 \right]$$

$$= 4 \times 10^6 \times \pi R^2 \times \frac{3}{4}$$

$$= 4 \times 10^6 \times \pi \times (4 \times 10^{-3})^2 \times \frac{3}{4} = 48\pi A.$$

Q17 (12)



$$I = \int J dA$$

$$= \int 10^6 \times 2\pi x dx$$

$$= 10^6 \times 2\pi \cdot x \cdot \frac{x^2}{2} \Bigg|_{\frac{r}{2}}^r$$

$$= \pi \times 10^6 \left[r^2 - \frac{r^2}{4} \right] = 12\pi$$

$$x = 12$$

Q18 (3)

$$R = \left(\frac{ma}{3} \right) + \left(\frac{a}{2m} \right)$$

$$\frac{dR}{dm} = \frac{a}{3} - \frac{a}{2m^2} = 0$$

$$\frac{a}{3} = \frac{a}{2m^2}$$

$$m^2 = \frac{3}{2}$$

$$m = \sqrt{\frac{3}{2}}$$

$$x = 3$$

Q19 (19)

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Hints and Solutions

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Using the conditions of a balanced wheat stone bridge and adding the end correction.

$$\frac{15}{(43+2)} = \frac{R}{(102-45)} \Rightarrow R = \frac{57}{45} \times 15$$

$$R = 19\Omega$$

Q20 (10)

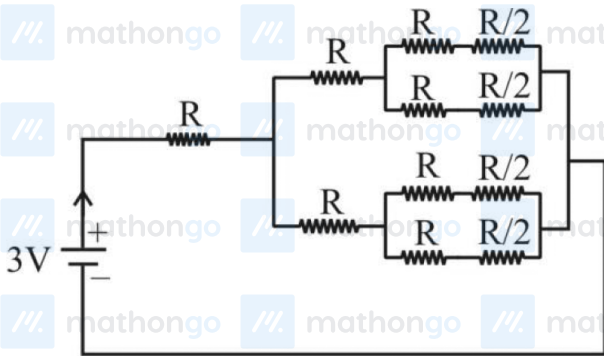
Using the condition of a balanced wheat stone bridge,

$$\Rightarrow \frac{R}{3} = \frac{4}{6} \Rightarrow R = 2\Omega$$

So the effective resistance of the circuit is

$$R_{eq} = \frac{6 \times 9}{6+9} = \frac{18}{5}\Omega$$

$$i = \frac{36}{R_{eq}} = 10A$$

Q21 (8)

$$R_{eq} = \frac{15R}{8} = \frac{15}{8}\Omega$$

$$I = \frac{3}{\frac{15}{8}} = \frac{8}{5}A$$

$$\therefore a = 8$$

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Q22 (15)

$$\frac{dQ}{dt} = i^2 R = \frac{V^2}{R} \text{ (we know)}$$

$$\Rightarrow \text{In 't' time, } \Delta Q = \left(\frac{V^2}{R} \right) t$$

Given that, (for same source, $v = \text{same}$)

$$Q_0 = \frac{V^2}{R_1} \times 20 = \frac{V^2}{R_2} \times 60 \dots (1)$$

$$\Rightarrow \boxed{R_2 = 3R_1} \dots (ii)$$

If they are connected in parallel then

$$R_{eq} = \frac{R_2 R_1}{R_1 + R_2} = \frac{3R_1 \cdot R_1}{3R_1 + R_1} = \left(\frac{3R_1}{4} \right)$$

To produce same heat, using equation ... (1)

$$Q_0 = \frac{V^2}{R_1} \times 20 = \frac{V^2}{\left(\frac{3R_1}{4} \right)} \times t$$

$$t = \frac{3 \times 20}{4} = 15 \text{ min}$$

Q23 (144)

$$1 = \rho \frac{\ell}{A}$$

$$1 = \frac{\rho \times 31.4}{\frac{\pi(2.4)^2}{4}}$$

$$\frac{\pi(2.4)^2}{4} = \rho \times 314$$

$$\frac{2.4 \times 2.4}{4} = \rho \times 10$$

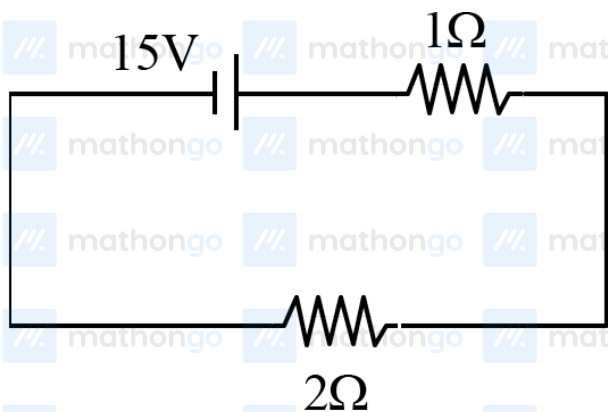
$$\frac{0.6 \times 2.4}{10} = \rho$$

$$\frac{1.44}{10} = \rho$$

$$0.144 = \rho$$

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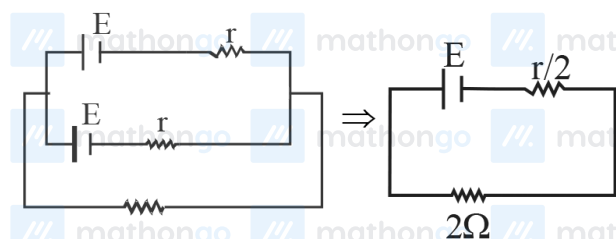
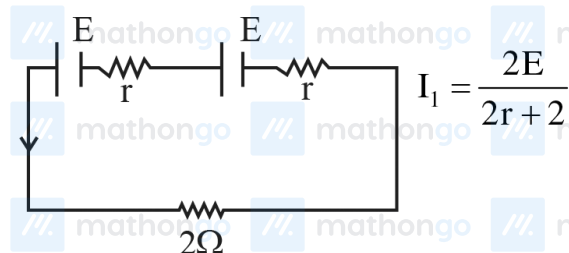
Q24 (10)



$$i = \frac{15}{3} = 5A$$

$$15 - 5(1) = 10 \text{ Volt}$$

Q25 (A)



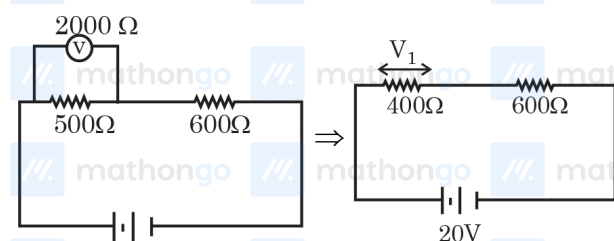
$$I_2 = \frac{E}{\frac{r}{2} + 2} = \frac{2E}{r + 4}$$

$$I_1 = I_2$$

$$2r + 2 = r + 4$$

$$2r - r = 2\Omega \Rightarrow r = 2\Omega$$

Q26 (8)



$$I = \frac{20}{1000} \text{ A}$$

$$V_1 = I \times 400 = \frac{20}{1000} \times 400$$

$$= 8 \text{ V}$$